

URL Shortener - Base 62 Conversion

1. Overview

Base conversion is another commonly used approach for generating short URLs.

Idea:

- Convert a **number (ID)** into a **Base62 string**
- Use this string as the **short URL**

2. Why Base 62?

Because we have **62 possible characters**:

None

0-9 → 10 characters

a-z → 26 characters

A-Z → 26 characters

Total = 62 characters

3. Character Mapping

Each number maps to a character:

None

0 → 0

1 → 1

...

9 → 9

10 → a

11 → b

...

35 → z

36 → A

...

61 → Z

4. Working Principle

Convert a base 10 number into **base 62 representation**

Example:

None

11157 (base 10)

5. Conversion Calculation

Break the number using powers of 62:

None

$$11157 = 2 \times 62^2 + 55 \times 62^1 + 59 \times 62^0$$

So:

None

[2, 55, 59]

	Remainder	Representation in base 62
62 11157	59	X
62 179	55	T
62 2	2	2
0		

6. Map to Base62 Characters

Using mapping:

None

2 → 2

55 → T

59 → X

7. Final Short URL

None

Base62 value = 2TX

Final shortened URL:

None

<https://tinyurl.com/2TX>

8. Key Insight

Instead of hashing:

- Generate a **unique numeric ID**
- Convert it into **Base62**

9. Advantages

- No collision (if ID is unique)
- Deterministic conversion
- Short and compact URLs
- Efficient and fast

10. Key Takeaways

- Base62 uses **62 characters for encoding**
- Converts numeric IDs into **short strings**
- Widely used in real-world URL shorteners
- Eliminates need for collision handling (unlike hashing)